Errata for the TME–EMT Wiki
(Art01 and related pages)

Compiled during formalization work in
`AlexKontorovich/PrimeNumberTheoremAnd`

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Abstract

This note collects discrepancies found while matching statements on the TME–EMT wiki against the cited source papers, in the course of formalizing those statements in Lean. Items are grouped by wiki page. Only high-confidence mismatches are listed; each entry gives the wiki text, the source, and a proposed correction.

How to read this note

Citations of the form "Art01" refer to Art01.html ("Explicit bounds on primes"). Page numbers for papers are from the versions available at the URLs recorded in the wiki bibliography.

1 Art01 — Explicit bounds on primes

1.1 Deléglise–Nicolas 2019, π_2 upper error constant

Wiki. Art01 quotes, for $x \geq 60173$,

$$\pi_2(x) - \left(\frac{x^3}{3\log x} + \frac{x^3}{9\log^2 x} + \frac{2x^3}{27\log^3 x}\right) \leq \frac{11181\, x^3}{648\log^4 x}.$$

Source. Deléglise–Nicolas, *The Landau function and the Riemann hypothesis*, arXiv:1907.07664, Corollary 2.7, display (2.21), uses the constant **1181** (not 11181). The same constant appears in the simplification leading to (2.22), as the coefficient 1181/216.

Proposed correction. Replace 11181 by 1181.

1.2 Platt–Trudgian 2021, ψ versus π

Wiki. Art01 attributes to Platt–Trudgian (2021) the bound

$$\left|\frac{\psi(x)-x}{x}\right| \leq 235\,(\log x)^{0.52}\exp\left(-\sqrt{\frac{\log x}{5.573412}}\right) \qquad (x \geq e^{2000}).$$

Source. Platt–Trudgian, *The Riemann hypothesis is true up to* $3 \cdot 10^{12}$, Table 1 at $X = 2000$ and Theorem 1 give a ψ-bound of the shape

$$\left|\frac{\psi(x)-x}{x}\right| \leq 411.4\left(\frac{\log x}{R}\right)^{1.52}\exp\bigl(-1.89\sqrt{\log x/R}\bigr), \qquad R = 5.573412.$$

The constants 235 and exponent 0.52 belong to the companion π-bound (Corollary 2), not to the ψ-bound.

Proposed correction. Quote the ψ-bound from Table 1 / Theorem 1, or explicitly relabel the displayed estimate as a π-bound and cite Corollary 2.

1.3 Johnston–Yang 2023, exponent on $\log x$

Wiki. Art01 quotes

$$\left|\frac{\psi(x)-x}{x}\right| \le 9.39\,(\log x)^{1.51}\exp\bigl(-0.8274\sqrt{\log x}\bigr) \qquad (x \ge 2).$$

Source. Johnston–Yang, Theorem 1.1, uses the exponent **1.515**.

Proposed correction. Replace 1.51 by 1.515.

1.4 De Koninck–Letendre 2020, missing k in $\log\log k$

Wiki. Art01, §4, writes

$$\sum_{i\le k} \log\log p_i \ge k\left(\log\log k + \frac{\log\log -3/2}{\log k}\right) \qquad (k \ge 6),$$

with a bare $\log\log$ in the numerator.

Source. De Koninck–Letendre, arXiv:1812.09950, Lemma 4.8 (4.21):

$$\sum_{i\le k} \log\log p_i \ge k\left(\log\log k + \frac{\log\log k - 3/2}{\log k}\right) \qquad (k \ge 6).$$

Proposed correction. Replace the bare $\log\log$ by $\log\log k$.

1.5 Büthe 2016, omitted π^* bound

Wiki. Art01, §1, under Büthe (2016), lists RH-conditional bounds for ψ, ϑ, and π only.

Source. Büthe also states

$$\bigl|\pi^*(x) - \mathrm{li}(x)\bigr| \le \tfrac{\sqrt{x}}{8\pi}\log x \qquad (x > 59),$$

under the same height hypothesis.

Proposed correction. Add the π^* bound to the displayed list.

1.6 Axler 2019, Mandl B_n mislabelled as $\sum p_i$

Wiki. Art01, §4, attributes to Axler (2019) the bounds

$$\sum_{i\le k} p_i \;\gtrless\; \frac{k^2}{4} + \frac{k^2}{4\log k} - \frac{k^2(\log\log k - c)}{4(\log k)^2}$$

(with $c = 2.9$ for the lower bound at $k \ge 6309751$, and $c = 4.42$ for the upper bound at $k \ge 256376$).

Source. Axler, *On the sum of the first n prime numbers*, arXiv:1606.06874, Theorems 1.6 and 1.7 bound Mandl's quantity

$$B_n = \frac{n\,p_n}{2} - \sum_{k\le n} p_k,$$

not $\sum p_k$. The actual prime-sum estimates are Theorems 1.8 and 1.9 (leading factor $n^2/(2\log n)$, different constants and ranges).

Proposed correction. Replace $\sum_{i\le k} p_i$ by B_k (and cite Theorems 1.6–1.7), or else quote Theorems 1.8–1.9 for the prime-sum bounds.

1.7 Cosmetic / typesetting (lower priority)

- Art01 opening paragraph: "van Mangold" → "von Mangoldt"; "direction verifications" → "direct verifications".
- Art01, Ramaré (2013) second display: unmatched parenthesis in the O^* term.
- Art01, Cipolla expansion: mixes ln and log.
- Art01, Kadiri (2008) table (§6): the entry under $\epsilon = 1$, $q_0 = 10^{25}$ is rendered as ".5116"; almost certainly "4.5116".

2 Art09 — Short intervals containing primes

2.1 Carneiro–Milinovich–Soundararajan 2019, interval orientation

Wiki. Art09 quotes, under RH and for $x \ge 4$, a prime in

$$\big]x - \tfrac{22}{25}\sqrt{x}\log x,\, x\big].$$

Source. Carneiro–Milinovich–Soundararajan, *Fourier optimization and prime gaps*, Comment. Math. Helv. 94 (2019); arXiv:1708.04122, Theorem 5:

$$\big[x,\, x + \tfrac{22}{25}\sqrt{x}\log x\big].$$

(The backward interval of that shape is Dudek's earlier result with constant $4/\pi$; Carneiro et al. improve the constant and reverse the orientation.)

Proposed correction. Replace the displayed interval by $[x,\, x + \frac{22}{25}\sqrt{x}\log x]$.

2.2 Trudgian 2016, Corollary 2 threshold

Wiki. Art09 quotes Trudgian (2016) for $x > 2{,}898{,}242$.

Source. Trudgian, *Updating the error term in the prime number theorem*, Ramanujan J. 39 (2016); arXiv:1401.2689, Corollary 2: $x \ge 2{,}898{,}239$.

Proposed correction. Replace 2,898,242 by 2,898,239, and use the non-strict threshold $x \ge \cdots$ as in the paper.

3 Art06 — Bounds on $|\zeta(s)|$

No high-confidence statement mismatches with the cited papers were found among the ζ-only results recorded in Art06 (Lehman, Cheng–Graham, Hiary, Backlund, Trudgian, Patel, Ford, Rosser, Hasanalizade–Shen–Wong, Ramaré, Delange). Year labels on some theorem heads ("2001", "2012") are citation noise only and do not affect the displayed inequalities.

Status in the Lean formalization

The corresponding statements in `PrimeNumberTheoremAnd/IEANTN/TMEEMT.lean` and `SecondarySummary.lean` have been aligned with the source papers for items 1.1, 1.2, 1.3, 1.4, 1.6, 2.1, and 2.2. Item 1.1 also required restoring the factor 2 in the π_2 main term $\frac{2x^3}{27\log^3 x}$, which Art01 already had correctly but which had been dropped in an earlier transcription into Lean. Item 1.6 was likewise present as a Lean transcription error (copied from the wiki); the formal statements now bound Mandl's B_n as in Theorems 1.6–1.7 of Axler. Items 2.1 and 2.2 were present both on Art09 and in Lean; both now follow the papers.